antropy: native multidimensional entropy and configurable complexity analysis

Methods, replicated synthetic studies, and resource-aware reproducibility

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Abstract

antropy: native multidimensional entropy and configurable complexity analysis provides a configurable framework for native spatial entropy, probability information, ordinal patterns, spectral distributions and trajectory structure. The implementation separates local spatial extraction from pair selection and estimator semantics. A common typed result preserves finite, undefined and infinite outcomes instead of collapsing them into missing numeric values. This manuscript accompanies antropy-hd 0.2.0.

The retained baseline contains 54 measured rows. The configured replicated study contains 150 fields and 1650 scalar estimates, including native HDSE, row-major flattened sample entropy and a fixed-alphabet entropy-complexity plane. Its field sizes are 48, 10 x 12, 6 x 7 x 8, with 5 independent generated trials per condition. Analytical work estimates report actual blocked-kernel schedules; empirical runtimes retain their host and repetition metadata. Source variables, figures and tables are generated from the stored reports and validated configuration.

The published HDSE construction motivates the local-template estimator, while the manuscript distinguishes source conventions, ambiguities and explicit extensions. The experiments are controlled synthetic demonstrations rather than a reproduction of the source paper's empirical classification results. Rendering uses the pinned shared research template's Pandoc, LaTeX, bibliography and real table-of-contents pipeline.

1 Introduction

Entropy is a family of questions about a specified representation, population and uncertainty model. A probability entropy asks about masses over an alphabet. A sample-template estimator asks how frequently similar local configurations remain similar after their representation is extended. Ordinal and spectral estimators construct different distributions. Their outputs require the input meaning, units, normalization and finite-sample convention to remain visible.

Native multidimensional data make these distinctions especially consequential. A spatial field has coordinate adjacency along several axes. Globally flattening it changes which values become adjacent in a one-dimensional embedding. Native HDSE extracts rectangular local blocks before their values are arranged for distance computation [Xia and Wang, 2026]. The mathematical estimator and the file representation are therefore distinct layers.

The repository name combines entropy and ant, reflecting interest in collective biological organization. No biological observations are included in the present study. The distribution and Python import use the separate names `antropy-hd` and `antropy_hd` to preserve compatibility with the unrelated AntroPy time-series library. The current registry exposes 19 orchestratable entries, while the numerical API also exposes reusable geometry, information and cost-estimation components.

The project has four connected aims: state estimator contracts precisely; preserve native geometry and explicit degeneracy; produce a replicated synthetic study with complete result coverage; and bind measured outputs to executable, resource-aware software and a configurable manuscript. The source repository is the configured public project.

2 Scope and related work

Shannon's discrete entropy defines information over a probability alphabet [Shannon, 1948]. Renyi and Tsallis generalize that functional while changing its order or extensivity conventions [Renyi, 1961, Tsallis, 1988]. Joint entropy, conditional entropy and mutual information retain the variable-axis meanings of probability tensors. Jensen-Shannon divergence compares distributions over

a shared alphabet [Lin, 1991]. These identities do not choose an amplitude discretization or establish a spatial embedding.

Approximate entropy and sample entropy make local-pattern continuation central [Pincus, 1991, Richman and Moorman, 2000]. HDSE extends local sample-template reasoning to higher-dimensional fields [Xia and Wang, 2026]. Fuzzy entropy replaces a discontinuous threshold indicator with a specified similarity kernel [Chen et al., 2007]. Multiscale entropy asks related questions after coarse graining [Costa et al., 2002]. Distribution entropy summarizes the empirical distribution of template distances [Li et al., 2015].

Permutation entropy constructs ordinal-pattern probabilities, with weighted permutation entropy retaining local amplitude variability [Bandt and Pompe, 2002, Fadlallah et al., 2013]. Spectral and trajectory decompositions provide additional representations [Rezek and Roberts, 1998, Broomhead and King, 1986]. The implementation uses a full multidimensional FFT periodogram and normalized singular values; neither convention is assumed identical to every spectral or SVD estimator in use.

Statistical complexity combines normalized entropy with disequilibrium. Here the Jensen-Shannon construction follows Rosso and colleagues and the broader family described by Martin and colleagues [Rosso et al., 2007, Martin et al., 2006]. Fixed-bin amplitude probabilities are a declared descriptive extension. This study does not infer Bandt-Pompe causality or diagnose noise versus chaos from those amplitudes.

The unified framework is an engineering and mathematical organization of these contracts. It does not assert that the estimators define one calibrated scalar ordering of complexity. Native HDSE versus flattened SampEn also changes feature count and extension geometry; that comparison is not a controlled intervention on flattening alone.

3 Methodology

3.1 Native spatial templates

Let x be a finite real-valued array with axis lengths $L = (L_1, \dots, L_d)$. An embedding specifies block shape m, delays δ, strides s and extension axes. For an origin i, the local template contains $x_{i+k\odot\delta}$ for $0 \le k_j < m_j$. Only this extracted block is flattened, in C order. Its feature count is $q = \prod_j m_j$.

Write m^+ for the extended shape. Common origins are chosen from the geometry supporting that extended block, so the paired population has

$$n = \prod_{j=1}^{d} \max\left(0, 1 + \left\lfloor \frac{L_j - 1 - (m_j^+ - 1)\delta_j}{s_j} \right\rfloor\right). \tag{1}$$

The numerical boundary rejects nonfinite, complex, boolean, string, object, empty and scalar inputs. Geometry and resource failures raise typed errors. Rectangular shapes, selected extension axes, delays and strides are explicit extensions of the paper's common isotropic window construction.

3.2 High-dimensional sample entropy

For paired base and extended templates, use the same admissible unordered origin pair set $\mathcal{P}$. With Chebyshev distance $D_\infty(u, v) = \max_k |u_k - v_k|$ and threshold $\tau = r\sigma_x$, define

$$B = \sum_{(i,j)\in\mathcal{P}} \mathbf{1}\{D_\infty(u_i, u_j) < \tau\}, \qquad A = \sum_{(i,j)\in\mathcal{P}} \mathbf{1}\{D_\infty(u_i^+, u_j^+) < \tau\}. \tag{2}$$

When $B > 0$ and $A > 0$,

$$\mathrm{HDSE}(x) = -\ln(A/B). \tag{3}$$

The implementation counts unordered pairs; doubling numerator and denominator recovers the corresponding ordered symmetric counting convention. The standard deviation is computed over the complete input, with population variance by default because the source does not specify the degrees-of-freedom convention. Strict thresholding follows the published HDSE definition [Xia and Wang, 2026].

The paper's prose requiring every origin coordinate to differ and its equations excluding only the identical origin tuple produce distinct populations. Ordinary tuple self-exclusion is the default. Coordinate exclusion is exposed as a documented sensitivity choice, along with Theiler exclusions and other distance norms. For HDSE, $B = 0$ is undefined; $B > 0, A = 0$ is infinite. A constant field has zero relative threshold, so strict matching yields no matches even when its amplitudes are identical.

The executable procedure is:

1. Parse the real array and immutable configuration.
2. Resolve common valid origins and guard template/pair storage.
3. Extract native base and extended windows on those origins.
4. Resolve global tolerance once from the declared standard deviation.
5. Count both representations with the same pair-selection policy.
6. Return value, status, populations, matching counts and effective parameters.

3.3 Related local-template estimators

One-dimensional SampEn uses the same paired engine with its separately declared boundary convention [Richman and Moorman, 2000]. Approximate entropy uses complete, separate origin populations at each block size and includes self matches. Its statistic is $\phi_m - \phi_{m+1}$, where $\phi_m = n_m^{-1} \sum_i \ln C_i^{(m)}$ [Pincus, 1991]. These populations and normalizers are not substituted into HDSE.

Fuzzy entropy uses the explicit kernel $K(D) = \exp[-(D/\tau)^p]$, optionally centering each template, and accumulates membership sums in the log domain [Chen et al., 2007]. Centering changes template semantics, so fuzzy continuation need not share all monotonicity properties of un-centered threshold counts. Nonfinite eligible distances or log weights raise a numerical-range error rather than becoming mathematical undefined estimates.

Multiscale analysis uses trimmed block means at the configured scale vector [Costa et al., 2002]. Original-input tolerance is preserved by default. Distribution entropy uses a specified histogram of pairwise distances; using all valid local origins, including the final valid origin omitted by the original DistEn construction, is explicitly an implementation extension [Li et al., 2015].

3.4 Probability, ordinal and decomposition estimators

For a declared probability alphabet of size M,

$$H_b(P) = -\sum_{a=1}^{M} p_a \log_b p_a, \tag{4}$$

with zero-mass terms contributing zero. Renyi entropy is $\log_b(\sum_a p_a^\alpha)/(1-\alpha)$, with explicit support-size, Shannon and min-entropy limits. Tsallis entropy is $(1-\sum_a p_a^\alpha)/(\alpha-1)$ with $k=1$ and a natural-log Shannon limit [Renyi, 1961, Tsallis, 1988]. Conditional entropy and mutual information use the named tensor-axis partitions, not the spatial-axis interpretation of an amplitude field.

Permutation entropy applies Shannon entropy to local ordinal-pattern probabilities; equal amplitudes are ordered stably by local C-order position [Bandt and Pompe, 2002]. Its weighted version uses local population variance [Fadlallah et al., 2013]. Spectral entropy normalizes the complete multidimensional FFT power distribution, retaining conjugate bins individually. SVD entropy normalizes singular values of the local trajectory matrix without squaring them [Rezek and Roberts, 1998, Broomhead and King, 1986]. Units, normalization and alphabet sizes accompany each result; singleton and zero-energy degeneracies are handled explicitly.

3.5 Jensen-Shannon statistical complexity

Let P_e be uniform on the complete configured alphabet and Δ_M a point mass on that same alphabet. The statistic is

$$C_{JS}(P) = \frac{H_b(P)}{\log_b M}\frac{J_b(P,P_e)}{J_b(\Delta_M,P_e)}, \qquad J_b(P,Q) = H_b\left(\frac{P+Q}{2}\right) - \frac{H_b(P)+H_b(Q)}{2}. \tag{5}$$

Zero-probability cells remain in the declared alphabet. A point mass has zero normalized entropy; a uniform distribution has zero disequilibrium. Their statistical complexities are therefore zero for different reasons [Rosso et al., 2007, Martin et al., 2006]. The final statistic is dimensionless. It complements entropy rather than acting as its renamed value.

The study's fixed-bin amplitude alphabet has $M-2$ interior intervals and two overflow cells, giving M total categories. Values below the lower bound or at least the upper bound remain in the corresponding overflow cell; empty cells remain in the probability vector. This symbolization is a declared descriptive extension, not a causal ordinal partition.

4 Experimental setup

4.1 Retained baseline

The original implementation demonstration is retained as a distinct dataset. Its shapes are 64, 16 x 16, 8 x 8 x 8, embedding shape 2, relative radii 0.2, 0.5, 1 and seed 20261009. It uses periodic, Gaussian and additive-mixture fields with period 8 and mixture amplitude 0.35. This smaller demonstration is not pooled with the replicated study.

4.2 Replicated Bernoulli-replacement design

The main study uses shapes 48, 10 x 12, 6 x 7 x 8, embedding shape 2, relative radii 1.5, 2, 2.5 and mixing probabilities 0, 0.25, 0.5, 0.75, 1. Each condition has 5 generated trials. The period is 12 and the root seed is 20261009. The complete consumed design is generated below from the actual stored plan.

Control	Configured value
schema version	1
seed	20261009
shapes	[[48], [10, 12], [6, 7, 8]]
mixing probabilities	[0.0, 0.25, 0.5, 0.75, 1.0]

Control	Configured value
controls	[“constant”, “checkerboard”, “gaussian”, “correlated”, “shuffled”]
replicates	5
radii	[1.5, 2.0, 2.5]
period	12.0
phase origin	1
embedding shape	2
gaussian sd	1.0
correlation steps	3
symbol range	[-4.0, 4.0]
symbol bins	16
max input elements	10000
max fields	200
max distance component visits	500000000
max templates	1000
max pairs	500000
max memory bytes	134217728
block size	64

The MIX construction follows the source paper's replacement form rather than the baseline's additive Gaussian mixture [Xia and Wang, 2026]. For coordinate u,

$$s(u) = \sum_{j=1}^{d} \sin(2\pi(u_j + u_0)/T), \qquad x_p(u) = (1 - \xi_{p,u})s(u) + \xi_{p,u}\eta_u, \tag{6}$$

where $u_0 = 1$ is the configured coordinate phase origin, ξ is Bernoulli with parameter p and η is uniform on $[-\sqrt{3}, \sqrt{3}]$. The periodic sum is not divided by $\sqrt{d}$. Common random numbers couple the mixing sweep within a trial, while the trial seed schedule separates generated fields. The design is bounded by explicit template, pair, input-element and workspace budgets.

Controls are constant, checkerboard, gaussian, correlated, shuffled. The Gaussian scale is 1; the configured correlation procedure uses 3 steps. Fixed amplitude symbolization declares 16 cells over -4.0, 4.0. Complete probability vectors include empty cells. Input meaning and symbolization policy are therefore part of the recorded configuration rather than inferred from a plot.

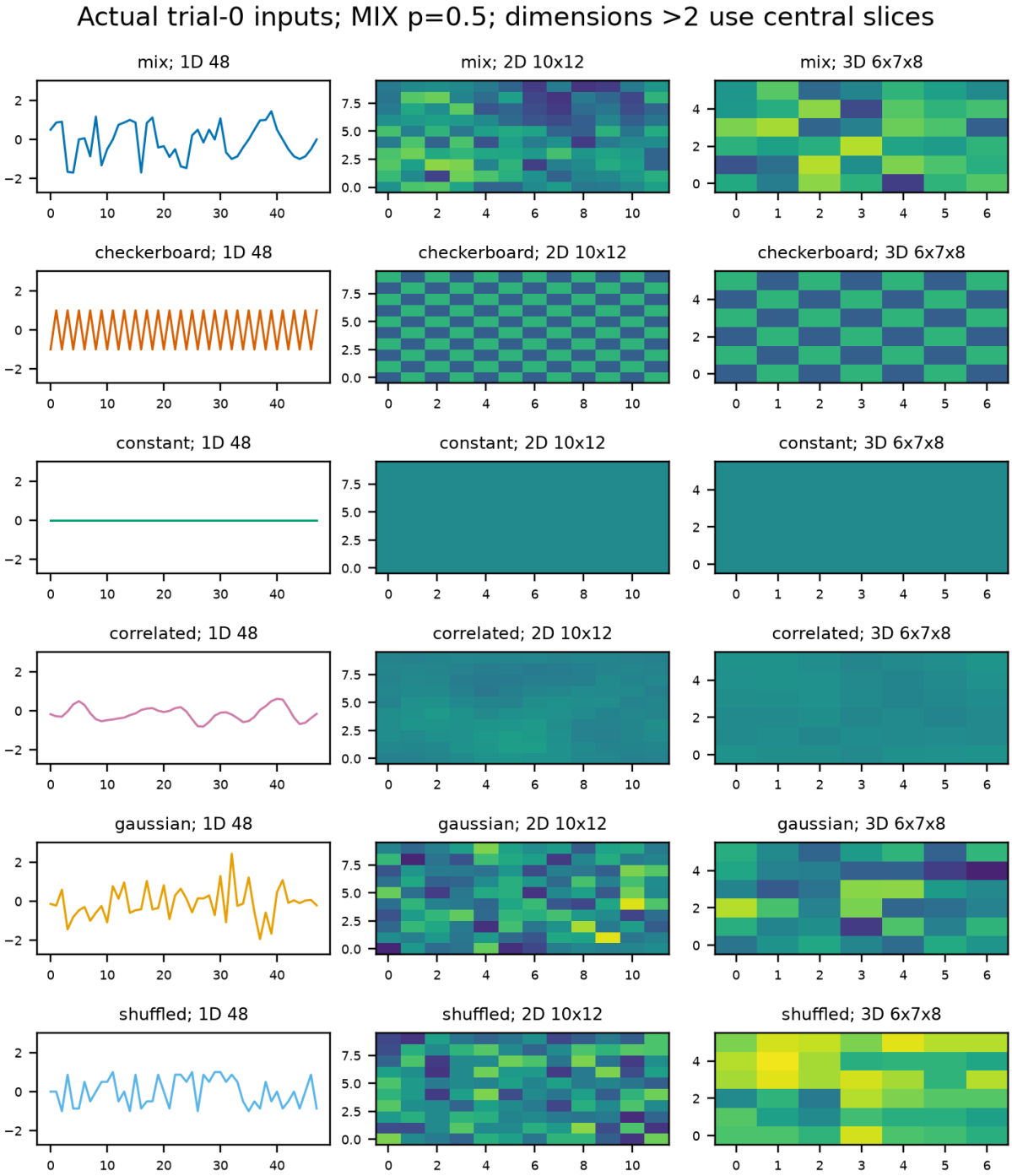


Figure 1: Illustrative generated fields from the stored synthetic design. Full inputs and seeds are preserved separately from their visual examples.

4.3 Geometry and repetition

Native HDSE and row-major flattened SampEn receive the same underlying amplitude field and relative-radius setting, but their local representations differ. A native block has m^d base features and an all-axis extension has $(m+1)^d$; the flattened one-dimensional continuation adds one feature. Origin populations and continuation dimensions also differ. These are comparison conditions, not an assumption that the estimators must agree.

Reported means and sample standard deviations condition on finite outcomes. Every summary separately retains total, finite, undefined and infinite counts. Generated trials are the unit of variation; overlapping template pairs are not treated as independent experimental replicates. No significance test or classification accuracy is inferred from this synthetic design. Deterministic conditions can produce identical fields across trial identities; their repeated values do not establish empirical uncertainty.

5 Computational complexity and resources

For n local templates with q components, explicit extraction requires $O(nq)$ component work and storage. Exhaustive pair comparison contributes $O(n^2q)$ distance-component work. Extending all axes replaces q by $q^+ = \prod_j (m_j + 1)$, so native-dimensionality changes feature count as well as the valid-origin population in eq. 1. Flattening changes both quantities.

The implementation processes bounded pair blocks. This controls workspace but does not remove quadratic total pair work. Its allocation-free estimator reports the actual scheduled distance cells, component visits, blocks, origin arrays and conservative guard bytes for the selected request. Distance work can include cells later excluded by pair masks; admissible pair counts and computed distance cells are distinct quantities. Exact scheduled counts, asymptotic size terms and host runtime observations retain separate interpretations.

Identical scheduled-work rows across radius settings are listed once below; the complete per-analysis estimates remain in the stored complexity report.

Here flat denotes row-major flattened observations; maximum and histogram denote the distance passes.

Input	Stage	n	q	Pairs	Visits	Guard (B)
6 x 7 x 8	base native	120	8	7140	86528	695936
6 x 7 x 8	extended native	120	27	7140	292032	695936
6 x 7 x 8	maximum native	210	8	21945	226848	683056
6 x 7 x 8	histogram native	210	8	21945	226848	683056
6 x 7 x 8	local native	210	8	0	0	85680
336	base flat	334	2	55611	132232	675488
336	extended flat	334	3	55611	198348	675488
10 x 12	base native	80	4	3160	21504	669056
10 x 12	extended native	80	9	3160	48384	669056
10 x 12	maximum native	99	4	4851	30244	669328
10 x 12	histogram native	99	4	4851	30244	669328
10 x 12	local native	99	4	0	0	20592
120	base flat	118	2	6903	20936	665120
120	extended flat	118	3	6903	31404	665120
48	base native	46	2	1035	4232	344864
48	extended native	46	3	1035	6348	344864
48	maximum native	47	2	1081	4418	363784
48	histogram native	47	2	1081	4418	363784
48	local native	47	2	0	0	7120
48	base flat	46	2	1035	4232	344864

Input	Stage	n	q	Pairs	Visits	Guard (B)
48	extended flat	46	3	1035	6348	344864

Guard bytes cover the controlled template and pair workspace. Caller-owned input arrays, file loading, standard-deviation temporaries, allocator overhead, vendor-library workspaces and process resident memory are outside that bound. The guard must reject infeasible requests; it is not a measurement of process RSS or a universal memory guarantee.

5.1 Empirical timing observations

The recorded environment is Python 3.14.7, NumPy 2.5.3, macOS-27.0.1-arm64-arm-64bit-Mach-O, machine arm64. The configured plan uses 1 warmups and 3 measured repetitions per request; block sizes are 16, 64.

Shape	Block	Median (s)	Sample SD (s)	Visits	Guard (B)
4 x 4 x 4	16	0.000233416	3.24125e-05	2240	16768
4 x 4 x 4	64	0.000179125	6.43296e-06	2240	16768
6 x 6	16	0.000105208	2.92041e-06	3328	46976
6 x 6	64	0.000101584	4.27311e-06	3328	46976
6 x 6 x 6	16	0.000730875	4.06458e-06	89600	64512
6 x 6 x 6	64	0.000358125	3.68909e-06	143360	678912
8 x 8 x 8	16	0.00741188	0.000489106	875840	110720
8 x 8 x 8	64	0.00221696	0.000139051	1041600	725120
10 x 10	16	0.000394875	2.604e-05	33280	52736
10 x 10	64	0.000279375	5.43607e-05	53248	667136
14 x 14	16	0.00151417	8.36652e-05	149760	62336
14 x 14	64	0.000536333	0.000197742	189696	676736
32	16	0.000118458	8.8355e-06	3380	46496
32	64	9.5167e-05	5.26603e-05	4500	149536
64	16	0.00025375	5.37063e-05	12020	48032
64	64	0.000103292	3.42341e-06	19220	622112
96	16	0.000437917	8.76008e-06	25780	49568
96	64	0.000261333	7.39915e-05	34580	663968

Timings are observed on the recorded environment and inputs. They do not prove an asymptotic theorem or promise portable latency. Block size can change workspace, call scheduling and runtime without changing the mathematical pair population or estimator value. Numerical validity, analytical cost and measured runtime are therefore checked and reported independently.

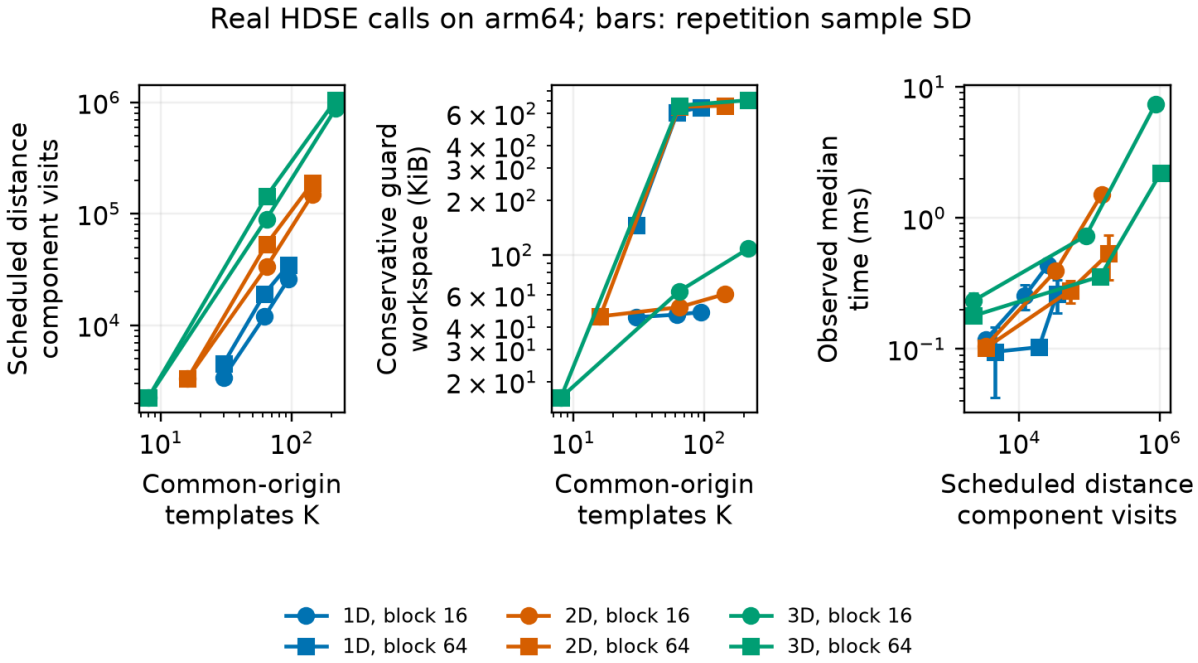


Figure 2: Measured runtime and configured deterministic work estimates. Host timings are diagnostic observations; exact scheduled counts and asymptotic terms have separate interpretations.

6 Computational results

6.1 Baseline outcomes

The baseline records 54 estimates: 45 finite, 3 undefined and 6 infinite. Its HDSE values are displayed directly from the stored CSV.

Shape	Field	r	HDSE (nats)	Status
64	periodic	0.2	0	finite
64	periodic	0.5	0.451985	finite
64	periodic	1	0.245482	finite
64	noise	0.2	1.4816	finite
64	noise	0.5	1.22536	finite
64	noise	1	0.680827	finite
64	mixed	0.2	2.70805	finite
64	mixed	0.5	1.02233	finite
64	mixed	1	0.488638	finite
16x16	periodic	0.2	0.262364	finite
16x16	periodic	0.5	1.07654	finite
16x16	periodic	1	0.765302	finite
16x16	noise	0.2	-	undefined
16x16	noise	0.5	-	infinite
16x16	noise	1	3.31381	finite
16x16	mixed	0.2	-	infinite
16x16	mixed	0.5	3.17109	finite
16x16	mixed	1	1.75318	finite
8x8x8	periodic	0.2	-	infinite
8x8x8	periodic	0.5	1.86322	finite
8x8x8	periodic	1	0.898039	finite
8x8x8	noise	0.2	-	undefined
8x8x8	noise	0.5	-	infinite
8x8x8	noise	1	-	infinite
8x8x8	mixed	0.2	-	undefined
8x8x8	mixed	0.5	-	infinite
8x8x8	mixed	1	4.71253	finite

6.2 Replicated native HDSE

The replicated report contains 150 fields and 1650 estimates: 1528 finite, 105 undefined and 17 infinite. The tables retain the finite conditional mean and sample SD, the finite fraction and the undefined/infinite counts (U/I). A dash indicates an unavailable finite mean or SD; it does not denote zero entropy.

Shape	p	r	Mean	SD	Finite/n	U/I
6x7x8	0	1.5	0.620344	0	5/5	0/0
6x7x8	0	2	0.334022	0	5/5	0/0
6x7x8	0	2.5	0.181913	0	5/5	0/0
6x7x8	0.25	1.5	3.71297	1.44226	5/5	0/0
6x7x8	0.25	2	2.21524	0.731486	5/5	0/0
6x7x8	0.25	2.5	1.18716	0.254853	5/5	0/0

Shape	p	r	Mean	SD	Finite/n	U/I
6x7x8	0.5	1.5	5.22817	0.343375	3/5	0/2
6x7x8	0.5	2	2.40793	0.402156	5/5	0/0
6x7x8	0.5	2.5	1.14763	0.136496	5/5	0/0
6x7x8	0.75	1.5	5.2898	0.184894	2/5	0/3
6x7x8	0.75	2	3.29457	0.316691	5/5	0/0
6x7x8	0.75	2.5	1.44975	0.191395	5/5	0/0
6x7x8	1	1.5	-	-	0/5	0/5
6x7x8	1	2	4.14886	0.305753	5/5	0/0
6x7x8	1	2.5	1.68958	0.112868	5/5	0/0
10x12	0	1.5	0.195222	0	5/5	0/0
10x12	0	2	0.0950618	0	5/5	0/0
10x12	0	2.5	0.0471768	0	5/5	0/0
10x12	0.25	1.5	0.845984	0.198192	5/5	0/0
10x12	0.25	2	0.412154	0.0775165	5/5	0/0
10x12	0.25	2.5	0.204453	0.0403457	5/5	0/0
10x12	0.5	1.5	1.39016	0.187022	5/5	0/0
10x12	0.5	2	0.637592	0.0698127	5/5	0/0
10x12	0.5	2.5	0.301306	0.0347635	5/5	0/0
10x12	0.75	1.5	1.84024	0.161139	5/5	0/0
10x12	0.75	2	0.977935	0.0408192	5/5	0/0
10x12	0.75	2.5	0.434886	0.0426958	5/5	0/0
10x12	1	1.5	2.00003	0.179567	5/5	0/0
10x12	1	2	1.06436	0.127857	5/5	0/0
10x12	1	2.5	0.425303	0.0656491	5/5	0/0
48	0	1.5	0.163193	0	5/5	0/0
48	0	2	0.0943835	0	5/5	0/0
48	0	2.5	0.0497278	0	5/5	0/0
48	0.25	1.5	0.34872	0.0451071	5/5	0/0
48	0.25	2	0.190946	0.0265122	5/5	0/0
48	0.25	2.5	0.0761487	0.0129646	5/5	0/0
48	0.5	1.5	0.395644	0.0426951	5/5	0/0
48	0.5	2	0.189375	0.0111342	5/5	0/0
48	0.5	2.5	0.0752926	0.00613628	5/5	0/0
48	0.75	1.5	0.387622	0.0279628	5/5	0/0
48	0.75	2	0.196965	0.023594	5/5	0/0
48	0.75	2.5	0.0802966	0.00841765	5/5	0/0
48	1	1.5	0.415829	0.022104	5/5	0/0
48	1	2	0.237607	0.0212663	5/5	0/0
48	1	2.5	0.0810962	0.015752	5/5	0/0

Matching probabilities provide another view of these outcomes. A positive base count followed by zero extended matches gives infinite continuation entropy; zero base matches give undefined entropy. Increasing native feature count and changing spatial geometry can therefore affect interpretability even when all input amplitudes are valid.

6.3 Structured controls and alternative representations

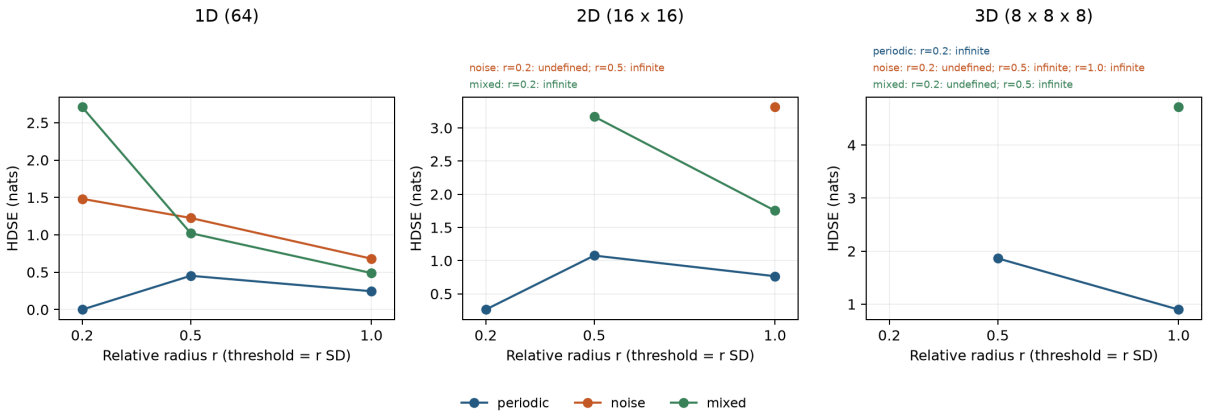


Figure 3: Baseline HDSE tolerance demonstration. Finite estimates are plotted; undefined and infinite estimates remain annotated and recorded in the table.

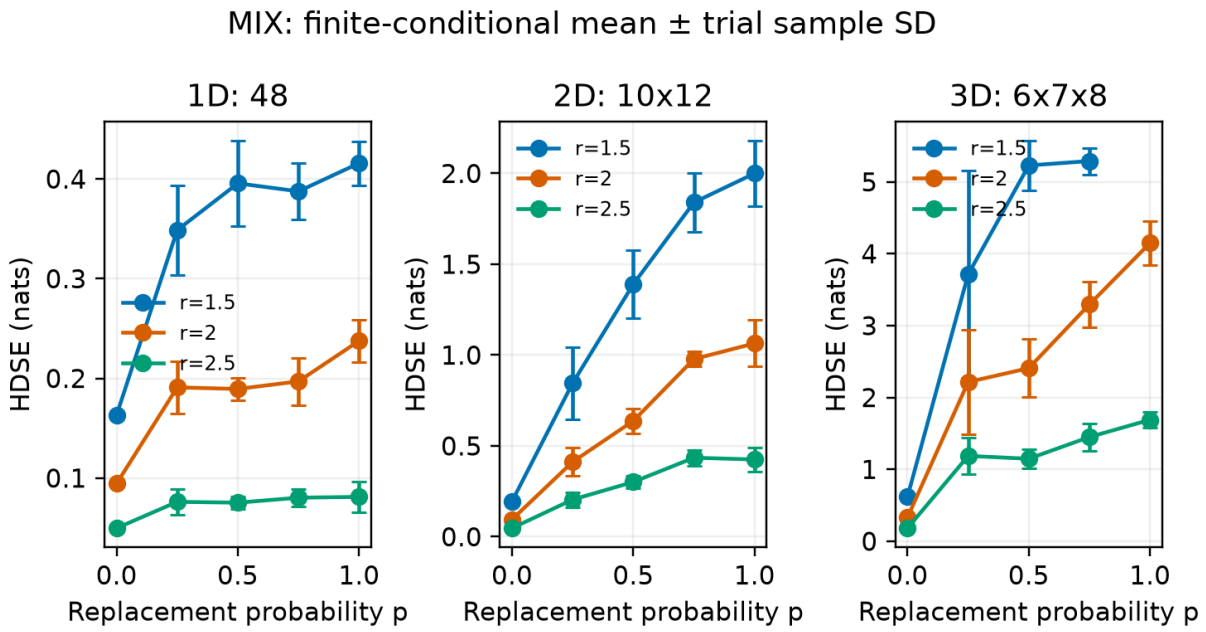


Figure 4: Native HDSE across Bernoulli replacement probabilities and configured radii. Finite conditional summaries are accompanied by outcome counts; variability is across generated fields, not template pairs.

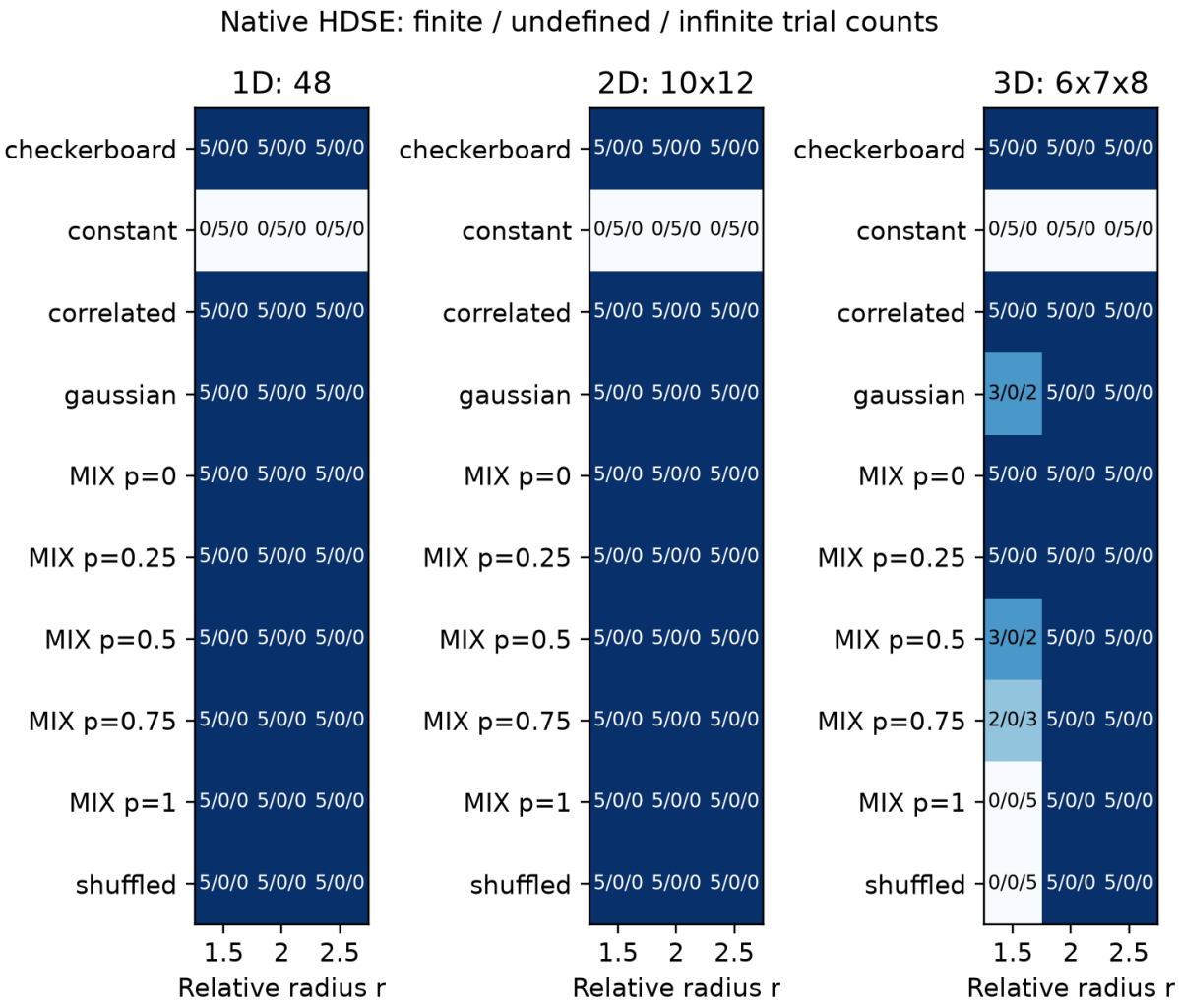


Figure 5: Finite, undefined and infinite outcomes across the replicated design. These statuses are retained rather than imputed as zero.

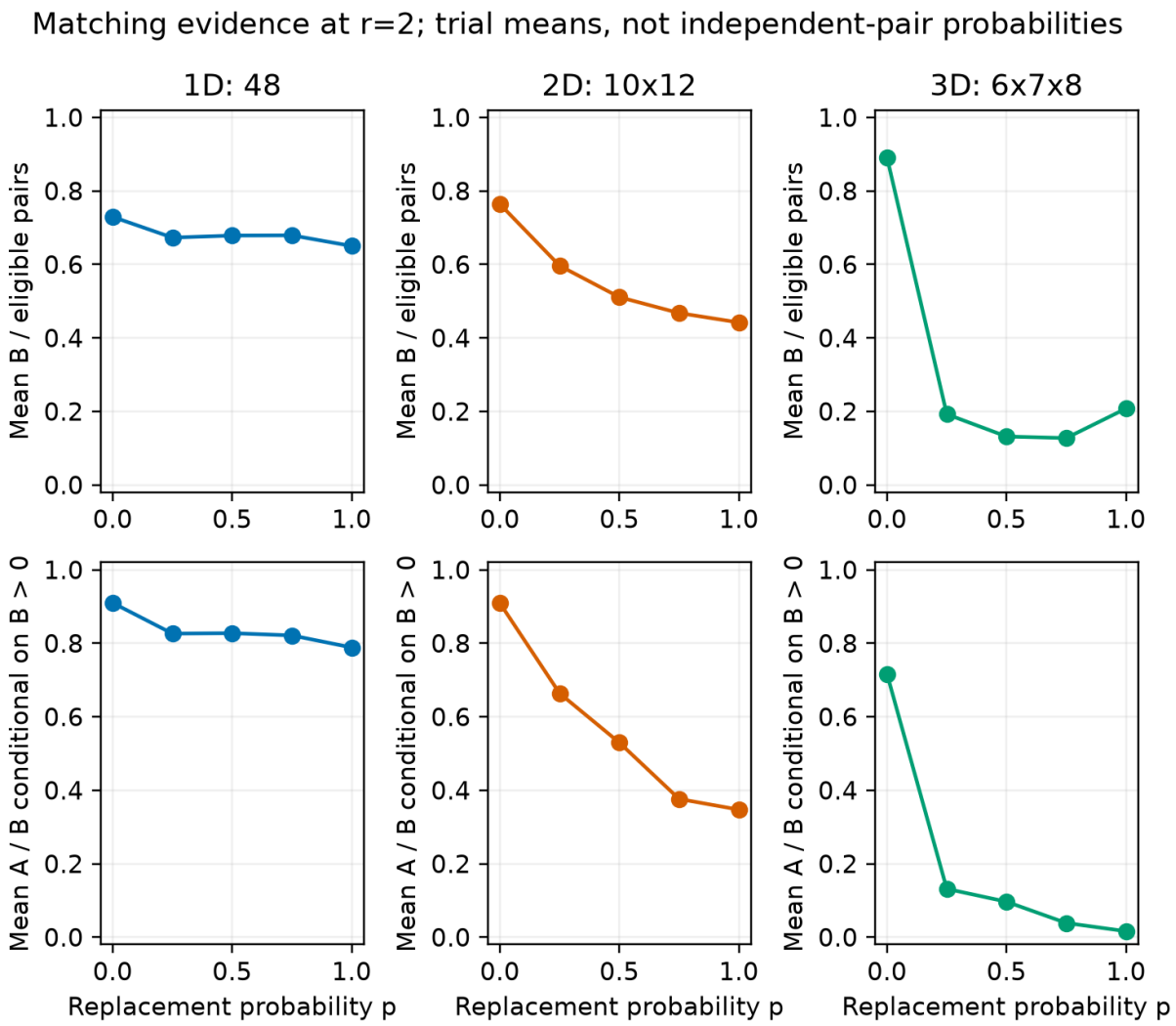


Figure 6: Base and extended template matching probabilities from observed pair counts. Sparse extended matches explain finite-sample degeneracy without reinterpreting it as zero entropy.

Shape	Control	r	Mean	SD	Finite/n	U/I
6x7x8	checkerboard	1.5	0	0	5/5	0/0
6x7x8	checkerboard	2	0	0	5/5	0/0
6x7x8	checkerboard	2.5	0	0	5/5	0/0
6x7x8	constant	1.5	-	-	0/5	5/0
6x7x8	constant	2	-	-	0/5	5/0
6x7x8	constant	2.5	-	-	0/5	5/0
6x7x8	correlated	1.5	2.56749	0.840728	5/5	0/0
6x7x8	correlated	2	1.37786	0.385547	5/5	0/0
6x7x8	correlated	2.5	0.753716	0.232607	5/5	0/0
6x7x8	gaussian	1.5	5.65724	0.495324	3/5	0/2
6x7x8	gaussian	2	3.3292	0.351848	5/5	0/0
6x7x8	gaussian	2.5	1.56662	0.232115	5/5	0/0
6x7x8	shuffled	1.5	-	-	0/5	0/5
6x7x8	shuffled	2	3.35688	0.471817	5/5	0/0
6x7x8	shuffled	2.5	1.76955	0.302802	5/5	0/0
10x12	checkerboard	1.5	0	0	5/5	0/0
10x12	checkerboard	2	0	0	5/5	0/0
10x12	checkerboard	2.5	0	0	5/5	0/0
10x12	constant	1.5	-	-	0/5	5/0
10x12	constant	2	-	-	0/5	5/0
10x12	constant	2.5	-	-	0/5	5/0
10x12	correlated	1.5	0.662661	0.109864	5/5	0/0
10x12	correlated	2	0.367965	0.0601135	5/5	0/0
10x12	correlated	2.5	0.185265	0.0324345	5/5	0/0
10x12	gaussian	1.5	1.74544	0.233531	5/5	0/0
10x12	gaussian	2	0.861448	0.128388	5/5	0/0
10x12	gaussian	2.5	0.400276	0.0465602	5/5	0/0
10x12	shuffled	1.5	1.79204	0.0637443	5/5	0/0
10x12	shuffled	2	0.850403	0.0587828	5/5	0/0
10x12	shuffled	2.5	0.409024	0.027914	5/5	0/0
48	checkerboard	1.5	0	0	5/5	0/0
48	checkerboard	2	0	0	5/5	0/0
48	checkerboard	2.5	0	0	5/5	0/0
48	constant	1.5	-	-	0/5	5/0
48	constant	2	-	-	0/5	5/0
48	constant	2.5	-	-	0/5	5/0
48	correlated	1.5	0.139492	0.0537352	5/5	0/0
48	correlated	2	0.0738259	0.0204956	5/5	0/0
48	correlated	2.5	0.0390499	0.0114501	5/5	0/0
48	gaussian	1.5	0.354011	0.0389952	5/5	0/0
48	gaussian	2	0.172839	0.026053	5/5	0/0
48	gaussian	2.5	0.0831918	0.0118893	5/5	0/0
48	shuffled	1.5	0.360859	0.0357175	5/5	0/0
48	shuffled	2	0.208973	0.018994	5/5	0/0
48	shuffled	2.5	0.0736555	0.00547319	5/5	0/0

The geometry comparison is read together with feature counts and populations. Neither a difference nor an agreement alone establishes estimator superiority. The observed tables are specific to the declared synthetic design, radius, finite sample and block geometry.

6.4 Related entropy and statistical complexity

The recorded normalization options make these related-measure summaries dimensionless. Display labels are DistEn: distribution entropy; PermEn: permutation; Spectral: spectral; Shannon: shannon; C-JS: statistical complexity.

Shape	p	Estimator	Finite mean	SD	Finite/n	U/I
6x7x8	0	DistEn	0.821272	0	5/5	0/0
6x7x8	0	PermEn	0.401562	0	5/5	0/0
6x7x8	0	Spectral	0.366254	0	5/5	0/0
6x7x8	0	Shannon	0.655986	0	5/5	0/0
6x7x8	0	C-JS	0.304078	0	5/5	0/0
6x7x8	0.25	DistEn	0.895451	0.00818597	5/5	0/0
6x7x8	0.25	PermEn	0.496046	0.00286137	5/5	0/0
6x7x8	0.25	Spectral	0.817511	0.0262881	5/5	0/0
6x7x8	0.25	Shannon	0.723031	0.00469153	5/5	0/0
6x7x8	0.25	C-JS	0.278564	0.00207836	5/5	0/0
6x7x8	0.5	DistEn	0.851999	0.00667078	5/5	0/0
6x7x8	0.5	PermEn	0.503431	0.000895235	5/5	0/0
6x7x8	0.5	Spectral	0.908797	0.0156529	5/5	0/0
6x7x8	0.5	Shannon	0.744093	0.00724335	5/5	0/0
6x7x8	0.5	C-JS	0.274515	0.00490519	5/5	0/0
6x7x8	0.75	DistEn	0.829532	0.00809795	5/5	0/0
6x7x8	0.75	PermEn	0.503852	0.000556784	5/5	0/0
6x7x8	0.75	Spectral	0.923803	0.00685935	5/5	0/0
6x7x8	0.75	Shannon	0.727915	0.00931895	5/5	0/0
6x7x8	0.75	C-JS	0.28245	0.00532371	5/5	0/0
6x7x8	1	DistEn	0.813712	0.00826181	5/5	0/0
6x7x8	1	PermEn	0.503478	0.000811645	5/5	0/0
6x7x8	1	Spectral	0.923819	0.00686466	5/5	0/0
6x7x8	1	Shannon	0.657322	0.00954592	5/5	0/0
6x7x8	1	C-JS	0.317542	0.00516416	5/5	0/0
10x12	0	DistEn	0.9211	0	5/5	0/0
10x12	0	PermEn	0.630036	0	5/5	0/0
10x12	0	Spectral	0.322933	0	5/5	0/0
10x12	0	Shannon	0.702175	0	5/5	0/0
10x12	0	C-JS	0.288771	0	5/5	0/0
10x12	0.25	DistEn	0.919696	0.0121944	5/5	0/0
10x12	0.25	PermEn	0.918879	0.0172518	5/5	0/0
10x12	0.25	Spectral	0.659491	0.0456326	5/5	0/0
10x12	0.25	Shannon	0.712075	0.00929196	5/5	0/0
10x12	0.25	C-JS	0.288816	0.00161396	5/5	0/0
10x12	0.5	DistEn	0.895524	0.0103245	5/5	0/0
10x12	0.5	PermEn	0.943309	0.0145696	5/5	0/0
10x12	0.5	Spectral	0.827952	0.0306717	5/5	0/0
10x12	0.5	Shannon	0.704415	0.00672185	5/5	0/0
10x12	0.5	C-JS	0.293462	0.00222378	5/5	0/0
10x12	0.75	DistEn	0.884852	0.00745327	5/5	0/0
10x12	0.75	PermEn	0.96377	0.00709848	5/5	0/0
10x12	0.75	Spectral	0.902698	0.0121331	5/5	0/0
10x12	0.75	Shannon	0.683502	0.0056531	5/5	0/0
10x12	0.75	C-JS	0.302502	0.00211954	5/5	0/0

Shape	p	Estimator	Finite mean	SD	Finite/n	U/I
10x12	1	DistEn	0.895472	0.00992491	5/5	0/0
10x12	1	PermEn	0.965971	0.00735781	5/5	0/0
10x12	1	Spectral	0.91021	0.0134747	5/5	0/0
10x12	1	Shannon	0.66164	0.00852925	5/5	0/0
10x12	1	C-JS	0.31495	0.00483384	5/5	0/0
48	0	DistEn	0.909918	0	5/5	0/0
48	0	PermEn	0.999673	0	5/5	0/0
48	0	Spectral	0.179052	0	5/5	0/0
48	0	Shannon	0.5	0	5/5	0/0
48	0	C-JS	0.331184	0	5/5	0/0
48	0.25	DistEn	0.915257	0.00752288	5/5	0/0
48	0.25	PermEn	0.993391	0.0035118	5/5	0/0
48	0.25	Spectral	0.777561	0.0341609	5/5	0/0
48	0.25	Shannon	0.578943	0.0354646	5/5	0/0
48	0.25	C-JS	0.325202	0.00666305	5/5	0/0
48	0.5	DistEn	0.929259	0.00851251	5/5	0/0
48	0.5	PermEn	0.986552	0.00983542	5/5	0/0
48	0.5	Spectral	0.845927	0.0531868	5/5	0/0
48	0.5	Shannon	0.613281	0.0207336	5/5	0/0
48	0.5	C-JS	0.322901	0.00458086	5/5	0/0
48	0.75	DistEn	0.948009	0.0133912	5/5	0/0
48	0.75	PermEn	0.993387	0.00574809	5/5	0/0
48	0.75	Spectral	0.882617	0.0293371	5/5	0/0
48	0.75	Shannon	0.629339	0.0270578	5/5	0/0
48	0.75	C-JS	0.320595	0.00573027	5/5	0/0
48	1	DistEn	0.958792	0.00697917	5/5	0/0
48	1	PermEn	0.999151	0.00116918	5/5	0/0
48	1	Spectral	0.880134	0.0274742	5/5	0/0
48	1	Shannon	0.63448	0.0272821	5/5	0/0
48	1	C-JS	0.318596	0.00497193	5/5	0/0

The entropy-complexity plane separates distributional uncertainty from distance to the uniform alphabet. A high normalized entropy and a high statistical complexity are different conditions. These fixed-amplitude-bin observations do not carry the causal ordinal interpretation of a Bandt-Pompe analysis.

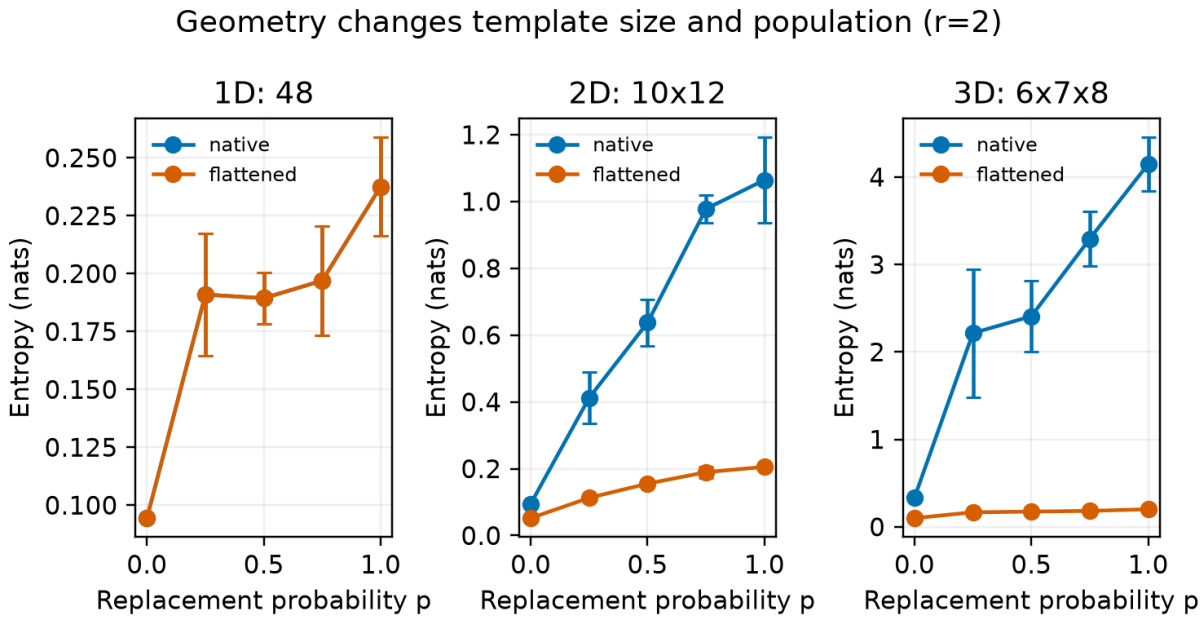


Figure 7: Native spatial HDSE and row-major flattened sample entropy. Their template feature counts and extension geometries differ; the comparison does not isolate flattening alone.

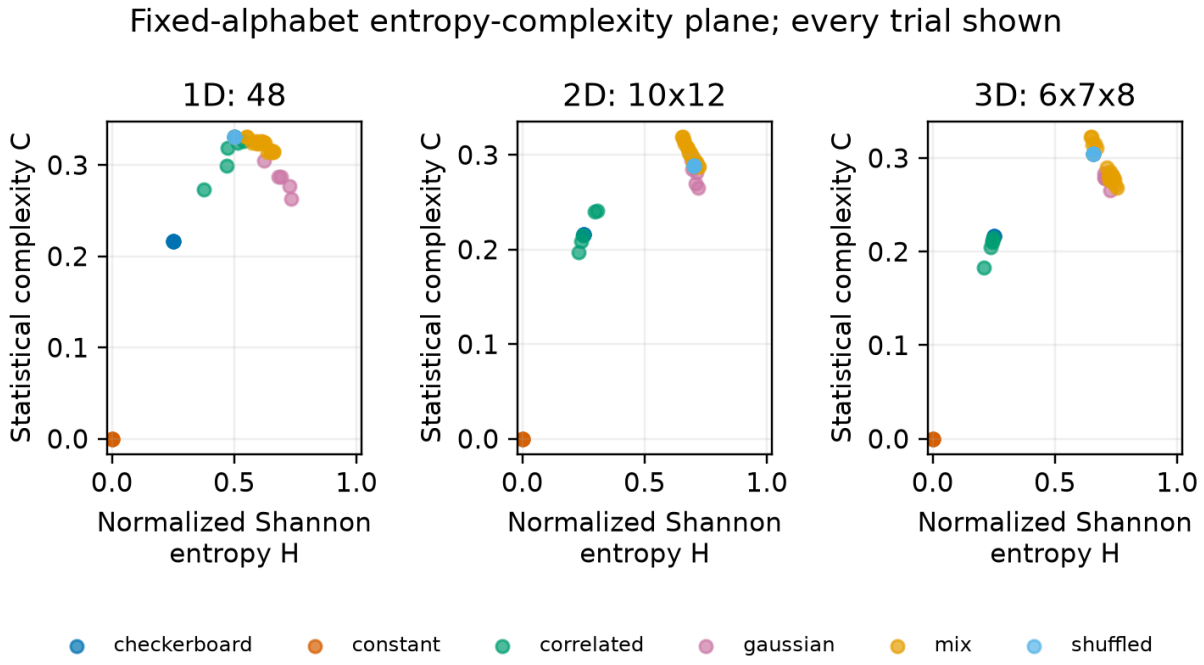


Figure 8: Normalized Shannon entropy and Jensen-Shannon statistical complexity of the complete configured fixed-bin amplitude alphabet. This is a descriptive distributional extension, not a causal ordinal classification.

7 Reproducibility and software architecture

7.1 Numerical and file boundaries

The numerical package separates array validation, geometry, blocked pair work and estimator-specific semantics. It does not import manuscript or plotting tooling. Immutable configurations reject unknown options. The registry binds the complete call signature, distinguishes observations from probability alphabets and retains effective parameters in every result. NPY loading disables pickle; CSV inputs retain their declared grid meaning. Strict JSON represents nonfinite estimates through explicit statuses and null values.

Tests use analytical examples, independently computed loop oracles, seeded arrays, real files and actual subprocesses. Deliberately invalid inputs and coherently altered artifact records are controls proving that checks can fail. Configured coverage floors are 90 percent for numerical source and 60 percent for research infrastructure.

The configured verification receipts report 613 passing tests, 95.6492 percent source coverage and 88.4959 percent infrastructure coverage. These are recorded runs, not a substitute for independent release review.

Receipt identities (SHA-256):

```
Source coverage: 482700cb812cba4b713afe1e47d1cadec4474b26c3ba8a9cc0631ba283e9987f
Infra coverage:  c3febbc9fb7a4ef63c2eda9cda2e8704f959960802a1597b5dad59cffcc352a5
Test log:        22b78552a93712b0018ed744b6519438fab29689ca6750dfb4a9186ae7cfee4c
```

7.2 Source-bound manuscript generation

The title, author metadata, version, design, measurements, tables and figure paths are sourced from the validated publication profile, project manifest and generated reports. Template variables must all resolve before rendering. The shared implementation is docxology/template, pinned at `d39f2478da2512b42ad961e98a5f3b854723f92e`, using the canonical `projects/templates/template_code_project/manuscript` conventions. Actual installed Python and Lua source files are authenticated against their installer RECORD hashes. Pandoc, Pandoc-crossref, BibTeX and the selected LaTeX engine produce mathematical typesetting, resolved references and a real TOC. No custom ReportLab manuscript fallback is used.

The inspectable resolved section tree, variable inventory, renderer source/tool receipt, generated TeX/TOC diagnostics and final PDF remain distinct artifacts. Figure staging uses exact-byte copies of producer outputs because the shared PDF renderer expects a sibling figure directory. Staging does not recompute or alter those figures.

7.3 Provenance bindings

```
Results CSV SHA-256: a4104429035874774f07cb2ec816a1cd656504435f245754d730e3bc06526727
Study CSV SHA-256: 493fd105d2410a1ac66c3bcf75e2a0313dc69e7b6ccd58625508454cc3ae57ce
Study report SHA-256: e77af9a7e5a55ddca73d9f779e55cf5e715d60e77b0fadd7c93f91f409f425be
```

The manifest binds actual input, configuration, code and output bytes. Stored file, canonical array, configuration, result and table identities remain exact. Synthetic design replay alone uses a declared, machine-epsilon-derived float64 envelope after exact shape and finite-value checks. This accommodates bounded platform roundoff without changing stored inputs, smoothing matching thresholds or relabeling stale results. The initial hosted failure and its adversarial regression controls are retained in the project record.

The supplied published PDF is not redistributed. No project DOI, journal acceptance, empirical source-paper replication or independent application validation is claimed by the software build.

Independent scientific and release review remains separate from these executable engineering checks.

8 Discussion and conclusions

Native geometry, input meaning and estimator convention are parts of a complexity analysis rather than preprocessing details that can be silently discarded. The framework makes those choices configurable and records their numerical consequences with counts and explicit statuses. The replicated study and computational records allow readers to examine tolerance, spatial representation, degeneracy and resource use together.

Finite conditional summaries must be interpreted alongside all outcome counts. The high-dimensional template feature population and finite origin geometry change together, while the native-versus-flattened comparison also changes continuation length. Stable ordinal ties, amplitude binning, spectral preprocessing and singular-value normalization are documented conventions. They remain application choices rather than universal definitions of complexity.

Exhaustive matching is quadratic in templates. Blocking bounds workspace, and the analytical estimator exposes the work schedule, but neither removes that growth. Host benchmarks describe observed execution on fixed inputs. New applications may require different data sizes, symbolization, boundary treatment or more efficient exact and approximate search methods, each with a new validation contract.

Ant collective behavior and other spatial scientific applications are future empirical domains, not findings of this synthetic study. Progress in those domains requires controlled measurements, appropriate null models, sensitivity analysis and independent review. The present contribution is a reproducible, typed methods implementation and source-bound research workflow through which those questions can be investigated.

References

Method definitions are attributed to their primary sources. The HDSE source, the local-template families, probability information, ordinal/decomposition methods and statistical-complexity family have separate citation records. Bibliographic entries are resolved by the shared template's native bibliography pipeline; source metadata do not import the template exemplar's publication DOI.

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